$$|\overrightarrow{AD}|^2 = \overrightarrow{AD}\cdot\overrightarrow{AD} = \left(\frac{\vec{b}+\vec{c}}{2}\right)\cdot\left(\frac{\vec{b}+\vec{c}}{2}\right) = \frac{1}{4}(|\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b}\cdot\vec{c}) \quad \ldots (1)$$

$$\text{Now, } \overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB} = \frac{\vec{b}+\vec{c}}{2} - \vec{b} = \frac{\vec{c}-\vec{b}}{2}.$$

Then, this gives, $|\overrightarrow{BD}|^2 = \overrightarrow{BD}\cdot\overrightarrow{BD} = \left(\frac{\vec{c}-\vec{b}}{2}\right)\cdot\left(\frac{\vec{c}-\vec{b}}{2}\right) = \frac{1}{4}(|\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b}\cdot\vec{c})$ ... (2)

Now, adding (1) and (2), we get

Therefore, $|\overrightarrow{AD}|^2 + |\overrightarrow{BD}|^2 = \frac{1}{4}(|\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b}\cdot\vec{c}) + \frac{1}{4}(|\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b}\cdot c) = \frac{1}{2}(|\vec{b}|^2 + |\vec{c}|^2)$

$\Rightarrow$ $|\overrightarrow{AD}|^2 + |\overrightarrow{BD}|^2 = \frac{1}{2}(|\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2)$.

Hence, $|\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 = 2(|\overrightarrow{AD}|^2 + |\overrightarrow{BD}|^2)$

Example 6.7

Prove by vector method that the perpendiculars (attitudes) from the vertices to the opposite sides of a triangle are concurrent.

Solution

Consider a triangle ABC in which the two altitudes AD and BE intersect at O. Let CO be produced to meet AB at F. We take O as the origin and let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$.

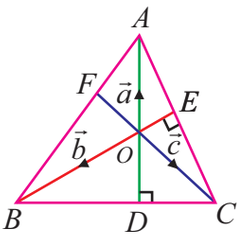


Fig. 6.12

Since $\overrightarrow{AD}$ is perpendicular to $\overrightarrow{BC}$, we have $\overrightarrow{OA}$ is perpendicular to $\overrightarrow{BC}$, and hence we get $\overrightarrow{OA}\cdot\overrightarrow{BC} = 0$. That is, $\vec{a}\cdot(\vec{c}-\vec{b}) = 0$, which means

$$\vec{a}\cdot\vec{c} - \vec{a}\cdot\vec{b} = 0. \quad \ldots (1)$$

Similarly, since $\overrightarrow{BE}$ is perpendicular to $\overrightarrow{CA}$, we have $\overrightarrow{OB}$ is perpendicular to $\overrightarrow{CA}$, and hence we get $\overrightarrow{OB}\cdot\overrightarrow{CA} = 0$. That is, $\vec{b}\cdot(\vec{a}-\vec{c}) = 0$, which means,

$$\vec{a}\cdot\vec{b} - \vec{b}\cdot\vec{c} = 0. \quad \ldots (2)$$

Adding equations (1) and (2), gives $\vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c} = 0$. That is, $\vec{c}\cdot(\vec{a}-\vec{b}) = 0$.

That is, $\overrightarrow{OC}\cdot\overrightarrow{BA} = 0$. Therefore, $\overrightarrow{BA}$ is perpendicular to $\overrightarrow{OC}$ which implies that $\overrightarrow{CF}$ is perpendicular to $\overrightarrow{AB}$. Hence, the perpendicular drawn from C to the side AB passes through O. Thus, the altitudes are concurrent.

Example 6.8

In triangle ABC, the points D, E, F are the midpoints of the sides BC, CA, and AB respectively. Using vector method, show that the area of ΔDEF is equal to $\frac{1}{4}$ (area of ΔABC).

Chapter 6 Vector Algebra.indd 228 31-01-2020 17:50:12